



Hydrodomus

Decentralizing hydrogen production

“Hydrogen generators for drones”



LA
FRENCH TECH
GRAND PARIS

The problem

Hydrogen drones fly farther, but they still wait for a delivery.



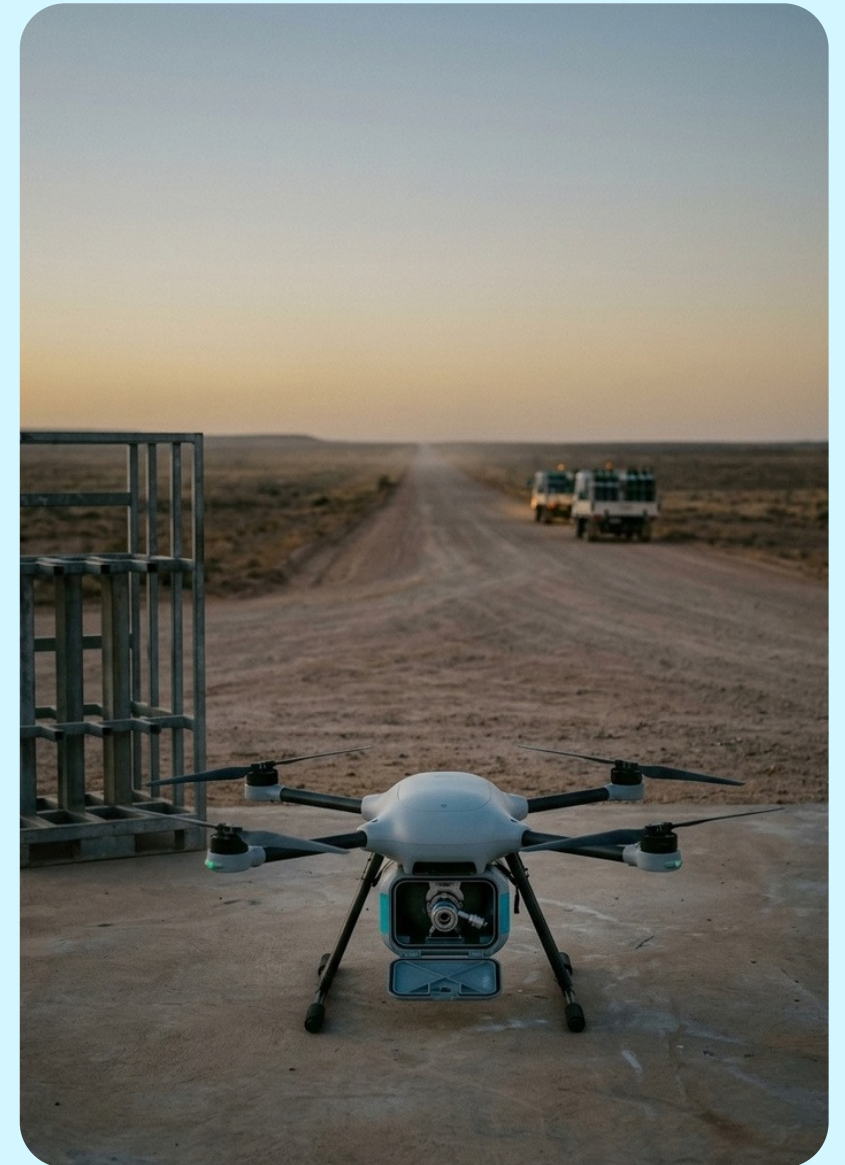
Made far away: hydrogen is produced at central plants, hundreds of kilometers from the front.



Delivered by convoy: full cylinders go in and empties come back, slowly and in the open.



Every sortie waits on it: when the delivery is late, the drone sits on the ground, however good its endurance is.



The solution

Make the hydrogen where the drone lands.



Hydrodomus concept :

“on-site hydrogen refueling node for drone fleets.”



It makes, compresses and dispenses hydrogen at the forward position, so nothing is shipped in.



The drone is filled on site, on demand, in under 5 minutes



Trailer-mounted, runs on a generator, grid, vehicle or solar, with fail-closed safety valves

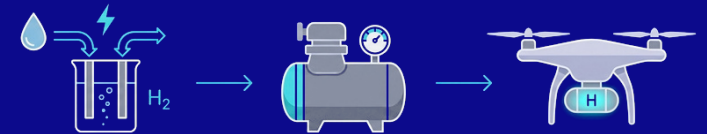


How it works



3 step process :

1. Generate hydrogen gas by electrolysis
2. Compress the gas at high pressure
3. Send it in the tank



Market validation

Hydrogen drones are already in service, funded or developed

Ukraine

A hydrogen ISR drone entered combat service in December 2025.

US Army

An ordering agreement now covers hydrogen generation equipment with the aircraft.

DIU and USMC

A hydrogen drone is on the Blue UAS Select list. HyTEC trailers are in testing with Marines.

France

RAPACE, an Air and Space Force research project, flew a military hydrogen drone in 2023.

NATO DIANA

Funded an on-demand field hydrogen generator, Phase 2.

Rheinmetall

Decentralized hydrogen for defense with ITM Power, announced April 2026.

Doosan

Sells 350 bar hydrogen drones commercially, refueled today by tank delivery.

Hydrodomus

1 of 3 target customers in discussion, meeting in October.

Market size

\$23B

Military drones, 2030

MarketsandMarkets

€17B

European military UAV spend, 2030

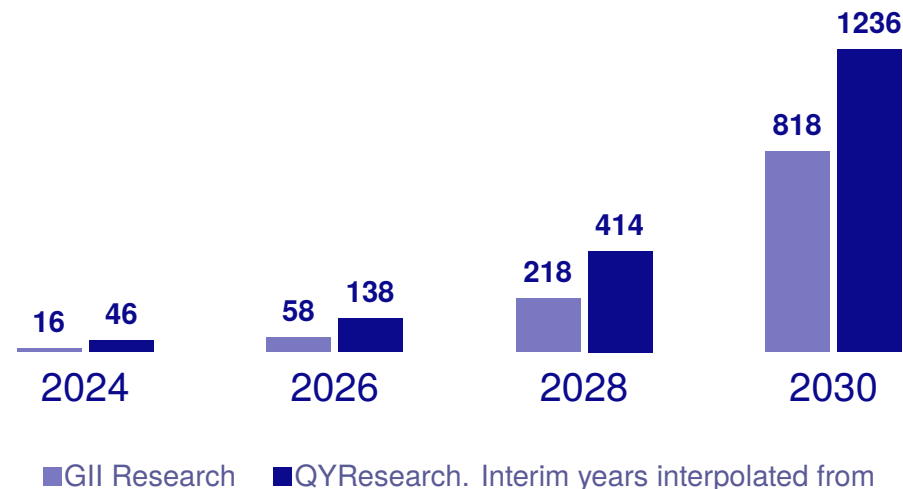
Drone Intelligence

\$1B

Hydrogen drones, 2030 (\$0.8B to \$1.2B)

GII Research, QYResearch

Hydrogen fuel-cell drone market, USD million



Benchmark

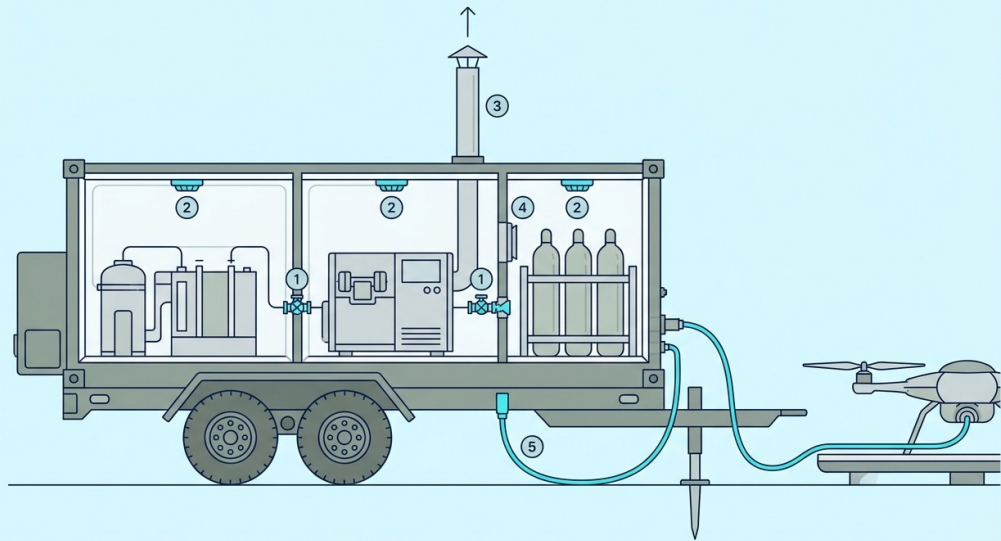
Solution	H ₂ on site	No consumable shipped	Output	Status
Tank delivery (Doosan)	No	No, cylinders	n/a	In use today
Doosan BHR-350 dispenser	No	No, needs H ₂ supplied	0 to 400 bar, 600 kg	On sale
MIT H-TaRP (aluminum)	Yes	No, aluminum	up to 20 fills/day	US Army demo
General Atomics (aluminum)	Yes	No, aluminum	not published	Army prototype
HyTEC (DIU)	Yes	Yes	about 2 kg/day	With US Marines
EHOSS (DIU, Pratt Miller)	Yes	Yes	over 20 kg/day	US program
H2-Field (H3 Dynamics)	Yes	Yes	not published	Selling since 2023
Hydrodomus TRN-10	Yes	Yes	10 to 25 kg/day	Design, bench proof

Closest peers are EHOSS, HyTEC and H2-Field. Our bet: a detachment-sized unit from sourced components, faster to a first pilot. That speed is what we have to prove.

From public information. “Not published” means we found no figure.

Safety

Safe by design: fail-closed, detected, vented, separated



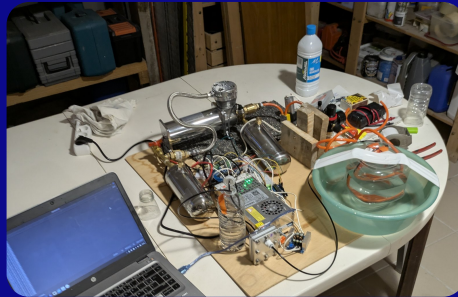
- 1 **Fail-closed isolation valves** between every zone. A fault or power loss isolates the system.
- 2 **Gas detection** in every bay, with automatic shutdown.
- 3 **Roof vent stack** sends any release up and away from people.
- 4 **Separate high-pressure bay** with a relief panel, apart from controls and operators.
- 5 **Grounding** of the chassis, the drone and the fill connection.

Standards we design to. Civil: ISO 19880, ISO 22734, PED 2014/68/EU, ATEX 2014/34/EU, Machinery 2006/42/EC, EMC 2014/30/EU. Military: MIL-STD-810, MIL-STD-461, MIL-STD-882. **Status:** design targets, nothing certified yet. Major components arrive certified from suppliers, and an independent hazard study is part of this round.

What works today, and what is bought



In-house bench prototype: PEM electrolysis, filtration, two-stage compression to 14 bar.



Works today

Bench-scale PEM electrolysis: 6 NL/h at 14 bar (about 13 g/day), multi-stage filtration, two-stage compression. This proves the core electrolysis know-how, not the product. The product target is roughly 770 times larger.

Bought, not scaled up

The production electrolyzer modules (two give about 10 kg/day) and the 350 bar compressor come from established industrial suppliers. The bench unit stays as the technology reference.

To do

Independent purity test, 350 bar integration, tank interface, ruggedization, certification, field trial. This is what the round funds.

Progress, and what is not done yet

1 of 3

target customers in discussion, meeting in October

11

supplier quotes in hand, from 20 requested

6 NL/h

working PEM bench electrolyzer

Feb 2027

speaker slot, Hydrogen & Fuel Cell Summit, Frankfurt

Done

- Pivoted the target from automotive to drone refueling and re-specified the product from scratch.
- Pilot bill of materials costed from supplier quotes.
- Founder full-time; Vienna Founders Residency; build base in Munich.

Not yet

- No revenue and no signed letter of intent: the October meeting is the step to get one.
- No 350 bar fill, no tank interface, no environmental testing.
- Pilot build starts once funded or committed by a customer.

Roadmap: four gates from quotes to a field pilot

1

Now

Turn quotes into firm offers, lock the pilot bill of materials, win the customer meeting.

Gate: signed LoI or pilot order

2

Bench demo

Sourced stack and compressor, fill fixture matched to the customer tank.

Gate: 350 bar fill into a real tank

3

Ruggedize

MIL-STD-810 and 461 testing, hazardous-area classification, transportable enclosure, with test labs.

Gate: environmental test report

4

Field pilot

A unit at a customer site, generating and filling under real conditions.

Gate: pilot data, follow-on order

Aim: show pilot progress and test data at the Hydrogen & Fuel Cell Summit, Frankfurt, February 2027.

Team



Yannick Durindel

Founder: business development
and engineering

MSc Hydrogen Engine, University of
Duisburg-Essen

ICAM engineering graduate, racecar engineering
(MUNER)

Former developer at Eyalights, consultant at mc2i

Built the bench prototype solo, full-time on
Hydrodomus

Advisors

David Holderbach

Hydrogen vehicles, founder of Hyvia.
Supplier and customer network.

Alysha Maria Lobo

Robotics and deep tech. Warm
customer introductions.

The ask

€750K

at €7.5M post-money (10%)

Funds, in order of priority:

- Pilot unit build and integration
- Environmental, EMC and safety testing
- Hazard studies and certification
- Runway to deliver the above

Yannick Durindel
yannick@hydrodomus.com

What the round delivers

1. A signed Lol or pilot order from the first customer
2. A 350 bar fill into a real drone tank
3. An environmental test report
4. A field pilot with real data

How you can help

Investors: introductions in defense tech, drones or deep tech.

Customers: introductions to drone operators and procurement.

Grants: leads on non-dilutive funding for pilots.

Appendix: sources and notes

- **Convoy casualties and fuel cost:** Military Writers / West Point analyses of Afghanistan fuel logistics (about 1 in 24 convoys with a casualty, about \$400 per gallon fully burdened on the most dangerous routes).
- **France RAPACE:** Ministry of the Armed Forces, CEA-Liten and AIN reporting on the Air and Space Force research centre (CREA) hydrogen drone, first flights 2023, with Atechsys as airframe maker.
- **Doosan DS30 and DT30:** Doosan Mobility Innovation data sheets; tank delivery with ReadyH2.
- **Hydrogen hybrid ISR drone in Ukrainian service:** public reporting, December 2025, with codification for wider fielding in January 2026.
- **DIU Blue UAS Select listing and US Army ordering agreement:** Heven Aerotech announcements, January and March 2026.
- **Decentralized hydrogen for defense:** Rheinmetall and ITM Power partnership announcement, April 2026.
- **NATO DIANA:** Phase 2 funding of an on-demand field hydrogen generator.
- **Market figures (not shown on slides):** hydrogen fuel-cell drone market from GII Research (\$15.6M in 2024 to \$818M in 2030) and QYResearch (\$26.8M in 2023 to \$1.24B in 2030); military drones from MarketsandMarkets (\$15.8B in 2025 to \$22.8B in 2030); European military UAV spend from Drone Intelligence. Analyst estimates differ widely by definition, so treat them as ranges.
- **Competitors:** DIU EHOSS (Pratt Miller), DIU HyTEC, H3 Dynamics H2-Field, MIT Lincoln Laboratory H-TaRP, General Atomics aluminum-water, Doosan BHR-350 and DS30 data sheets.
- **Our numbers:** bench figures are measured (6 NL/h at 14 bar). TRN-10 figures are design targets. Power figures assume about 55 kWh per kg at the stack and typical generator consumption. Supplier prices are from quotes, not final offers.